Purity Priced: The Registration-Costs-Bonds Family

Working paper v0.1, Project secret3, Row A, line W3, August 11, 2026

1. The claim

To register is to build, and building stands on bonds. The purity fact itself, that a rewriting cosmos has no side channels and a meeting's output is everything the meeting produces, is a design truth of the framework, stated as such since the concept note. What this campaign compiles is that fact's countable shadow, a family of six results on the bond meter: the bondless are silent; what has no structure goes nowhere and writes nothing; any change at all presupposes at least one standing bond; the Kestrel's bill is exact, two bonds erased and a ledger discarded whole; and the Starling's entire growth is one copy of the shared ground, growth as copying and never as minting, the bond-level twin of W2′ s provenance.

2. The theorems as checked

Module Purity, the fifteenth of the package, checked in seconds. The meter: bonds counts application nodes, one per meeting-in-standing. The family:

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zeroBondsAtomic     : bonds t = 0 -> AtomicT t
atomsAreSilent      : AtomicT t -> Step t u -> Void
bondlessWalksNowhere: bonds t = 0 -> Steps t u -> u = t
registrationCosts   : Steps t u -> Not (u = t) -> Not (bonds t = 0)
kFiringBill         : bonds (App (App CK a) b) = 2 + (bonds a + bonds b)
sFiringBill         : bonds (App (App a c) (App b c))
                    = bonds (App (App (App CS a) b) c) + bonds c
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The first four form the silence ladder: no bonds, then atomic; atomic, then no move; no move, then the only walk is the empty walk; and therefore any real change certifies at least one bond standing ready before it. The two bills price the moves themselves, with sFiringBill carried by a spine ledger, a result ledger, and one pure ground shuffle, and its reading is the campaign's center: every bond the Starling adds is a copy of a bond the ground already carried.

3. The purity reading, assembled

Put the family beside the delivered edges and the identity the concept note promised becomes a chain of receipts. The crown, S K K, returns its partner untouched, skIdentity; a partner returned untouched is a walk ending where the record would have to differ, and registrationCosts says difference presupposes structure; the crown's own three bonds fire and vanish in the return, leaving the output that is, by design, everything; so perfect gentleness and perfect blankness are the same ledger read twice. In a pure world, to know is to touch, and the theorem family now prices the touch: at least one bond to act at all, two erased per forgetting, one ground copied per sharing.

4. Readings, both at full strength

The offered reading, for chapters eleven and twelve: observation without disturbance is observation without record, now with the cost table attached; the witness of the traditions, registering without touching, is priced out of form not rhetorically but arithmetically. The sober reading, printed whole: these are elementary structural counts on a first-order syntax, their compilation is

bookkeeping, and the words record, silence, and gentleness are chosen; a deflationist reads the same table as inventory management. Both stand; the chair is the reader's.

5. Service and status

W3 closes as delivered, packaged. Line W now reads: W1 delivered, W2 delivered, W3 delivered, W4 tier B open by design. Part Three's silhouette stands on compiled edges alone except where a termination certificate is honestly named. Package v0.5, fifteen modules; the fast lane's tally tonight: four campaigns, four papers, every theorem in seconds.